

1. Log Analyzer

```
hecker@Windows: ~/Project_Os/Week_4
hecker@Windows:~/Project_Os/Week_4$ echo -e "10:00 serviceA error\n10:05 serviceB ok\n10:10 serviceA error\n10:15 serviceC error\n10:20 serviceA error\n10:25 serviceB ok" > log.txt
hecker@Windows:~/Project_Os/Week_4$ grep error log.txt
10:00 serviceA error
10:10 serviceA error
10:15 serviceC error
10:20 serviceA error
hecker@Windows:~/Project_Os/Week_4$ grep error log.txt | awk '{print $2}' | sort | uniq -c
  3 serviceA
  1 serviceC
hecker@Windows:~/Project_Os/Week_4$ grep error log.txt | awk '{print $2}' | sort | uniq -c | sort -nr | head -5
  3 serviceA
  1 serviceC
hecker@Windows:~/Project_Os/Week_4$
```

2. Network Debugging

```
hecker@Windows: ~/Project_Os/Week_4
hecker@Windows:~/Project_Os/Week_4$ ping -c 3 google.com
PING google.com(del03s05-in-x0e.1e100.net (2404:6800:4002:803::200e)) 56 data bytes
64 bytes from del03s05-in-x0e.1e100.net (2404:6800:4002:803::200e): icmp_seq=1 ttl=118 time=10.4 ms
64 bytes from del03s05-in-x0e.1e100.net (2404:6800:4002:803::200e): icmp_seq=2 ttl=118 time=13.9 ms
64 bytes from del03s05-in-x0e.1e100.net (2404:6800:4002:803::200e): icmp_seq=3 ttl=118 time=13.2 ms

--- google.com ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2001ms
rtt min/avg/max/mdev = 10.391/12.483/13.879/1.506 ms
hecker@Windows:~/Project_Os/Week_4$ curl -I https://google.com
HTTP/2 301
location: https://www.google.com/
content-type: text/html; charset=UTF-8
content-security-policy-report-only: object-src 'none';base-uri 'self';script-src 'nonce-EK13f5D-Fzi4ufSgpC0-rw' 'strict-dynamic' 'report-sample' 'unsafe-eval' 'unsafe-inline' https: http;report-uri https://csp.withgoogle.com/csp/gws/other-hp
reporting-endpoints: default="//www.google.com/htpservice/retry/jserror?ei=YqXKad-LDa2w4-EPmYb_yAc6cad=crash&error=Page%20Crash6jsel=1&bver=2410&dpf=jsj0LLRsoI7REJE82RmMyG0LRgu--lcETVHW13Lhxxx"
date: Mon, 30 Mar 2026 16:31:30 GMT
expires: Wed, 29 Apr 2026 16:31:30 GMT
cache-control: public, max-age=2592000
server: gws
content-length: 220
x-xss-protection: 0
x-frame-options: SAMEORIGIN
alt-svc: h3=":443"; ma=2592000,h3-29=":443"; ma=2592000

hecker@Windows:~/Project_Os/Week_4$ ss -tuln
Netid      State      Recv-Q     Send-Q               Local Address:Port      Peer Address:Port      Process
udp        UNCONN     0           0                   127.0.0.0.53%lo:53      0.0.0.0:*
udp        UNCONN     0           0                   127.0.0.1:323          0.0.0.0:*
udp        UNCONN     0           0                   0.0.0.0:50188          0.0.0.0:*
udp        UNCONN     0           0                   224.0.0.251:5353       0.0.0.0:*
udp        UNCONN     0           0                   224.0.0.251:5353       0.0.0.0:*
udp        UNCONN     0           0                   224.0.0.251:5353       0.0.0.0:*
udp        UNCONN     0           0                   224.0.0.251:5353       0.0.0.0:*
udp        UNCONN     0           0                   0.0.0.0:5353           0.0.0.0:*
```

```
hecker@Windows: ~/Project_Os/Week_4
date: Mon, 30 Mar 2026 16:31:30 GMT
expires: Wed, 29 Apr 2026 16:31:30 GMT
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content-length: 220
x-xss-protection: 0
x-frame-options: SAMEORIGIN
alt-svc: h3=":443"; ma=2592000,h3-29=":443"; ma=2592000

hecker@Windows:~/Project_Os/Week_4$ ss -tln
Netid      State      Recv-Q     Send-Q      Local Address:Port      Peer Address:Port      Process
udp        UNCONN     0           0            127.0.0.53%lo:53        0.0.0.0:*
udp        UNCONN     0           0            127.0.0.1:323          0.0.0.0:*
udp        UNCONN     0           0            0.0.0.0:50188          0.0.0.0:*
udp        UNCONN     0           0            224.0.0.251:5353       0.0.0.0:*
udp        UNCONN     0           0            224.0.0.251:5353       0.0.0.0:*
udp        UNCONN     0           0            224.0.0.251:5353       0.0.0.0:*
udp        UNCONN     0           0            224.0.0.251:5353       0.0.0.0:*
udp        UNCONN     0           0            0.0.0.0:5353           0.0.0.0:*
udp        UNCONN     0           0            [::]:56475             [::]:*
udp        UNCONN     0           0            [::1]:323              [::]:*
udp        UNCONN     0           0            [fe80::d727:82ff:6dad:7d1c]::wlp6s0:546 [::]:*
udp        UNCONN     0           0            [::]:5353               [::]:*
tcp        LISTEN     0           4096         127.0.0.1:9050          0.0.0.0:*
tcp        LISTEN     0           128          127.0.0.1:631           0.0.0.0:*
tcp        LISTEN     0           4096         127.0.0.53%lo:53        0.0.0.0:*
tcp        LISTEN     0           4096         127.0.0.1:39429         0.0.0.0:*
tcp        LISTEN     0           128          [::1]:631               [::]:*

hecker@Windows:~/Project_Os/Week_4$ ss -tln | grep LISTEN
tcp        LISTEN     0           4096         127.0.0.1:9050          0.0.0.0:*
tcp        LISTEN     0           128          127.0.0.1:631           0.0.0.0:*
tcp        LISTEN     0           4096         127.0.0.53%lo:53        0.0.0.0:*
tcp        LISTEN     0           4096         127.0.0.1:39429         0.0.0.0:*
tcp        LISTEN     0           128          [::1]:631               [::]:*
```

3.Data Extraction

```
hecker@Windows: ~/Project_Os/Week_4
hecker@Windows:~/Project_Os/Week_4$ echo -e "1,apple,10\n2,banana,20\n3,apple,15\n4,banana,25" > data.csv
hecker@Windows:~/Project_Os/Week_4$ cat -d',' -f2 data.csv
apple
banana
apple
banana
hecker@Windows:~/Project_Os/Week_4$ grep apple data.csv
1,apple,10
3,apple,15
hecker@Windows:~/Project_Os/Week_4$ grep apple data.csv | cut -d',' -f3 | awk '{sum += $1} END {print sum}'
25
hecker@Windows:~/Project_Os/Week_4$
```